CEBT II REACTION ENGINEERING

Recommended text: O. Levenspiel "Chemical Reaction Engineering", 3rd edition, Wiley, 1998.

1. INTRODUCTION

- Heterogeneous reactors have more than one phase present.
  - Mass transfer between (or within – Mixedness, CEBT 1B Reaction Engineering 1) the phases may be important as well as chemical kinetics.
- This course will concentrate on heterogeneous reactors in which one phase is a solid.
  - The solid may be a reagent e.g. gas-solid and liquid-solid reactions (Section 2)
  - The solid may be a catalyst rather than a reagent; solid catalysts are involved in the manufacture of >90% of chemical products, in part because of easy separation of products from the catalyst. Reaction engineering with a solid catalyst needs consideration of
    - types of catalyst and reactor (Section 3)
    - adsorption on surfaces (Section 4)
    - reaction on surfaces (Section 5)
    - reaction in porous solids (Section 6)
  - Professor Lapkin will cover gas-liquid reactors in lectures 13 and 14 (Section 7), including a short continually assessed computational project.
  - A particularly important class of bio-reactor is the fermenter, in which the reaction rate is often limited by transfer of oxygen from gas to liquid phase (Section 8).
- Before that, we shall revise some of the key concepts from CEBT 1B Reaction Engineering, which will be required for this course.

1.1 Rate of reaction

- For homogeneous reactions, we defined the rate of reaction for species A to be
  r_A the moles of A made per second per unit volume
- For heterogeneous reactions, the rate of reaction may be defined based on
  - Volume of reactor (as for homogeneous reactions)
  - Surface area of solid (if reaction is only taking place at the surface of a solid)
  - Mass of solid catalyst (if reaction is taking place within a porous solid)
- We shall use the symbols r_V, r_{SA} and r_M to distinguish between these different cases if there is any ambiguity.

- The phrase "rate of reaction" is also often used to denote the total moles per second made.
- Always look at the units specified.

- When the reaction takes place on a solid catalyst, another term frequently used to describe reaction rate is the

  TURNOVER FREQUENCY, N = number of molecules reacting per second per active site

  - This is simply a rate of reaction, but using a basis of "active sites".

- When there is more than one reagent, the rates of reaction for different species are related to each other by the reaction stoichiometric coefficients ν_i

$$\frac{r_A}{\nu_A} = \frac{r_B}{\nu_B}$$

  - We shall label the rate of reaction (for unspecified species) to be $r = r_i/\nu_i$

- The rate of reaction is given by a rate law.
- Example rate laws for homogeneous reactions are

$r = kC_A$	First-order kinetics with respect to A
$r = kC_A^2$	Second-order kinetics with respect to A
$r = \frac{kC_A}{K_M + C_A}$	Michaelis-Menten kinetics (typical of enzymes)
$r = kC_AC_B$	First-order kinetics with respect to both A and B (second order overall)

  - Example: reaction 2A → 3B with second-order kinetics with respect to A:

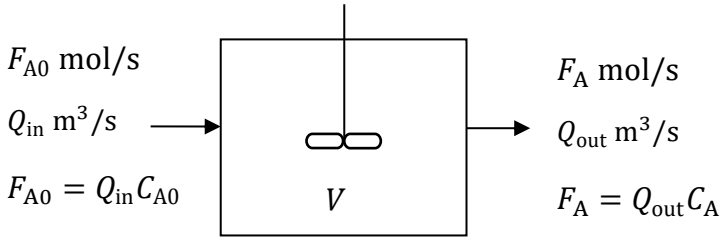
$$r = \frac{-r_A}{2} = \frac{r_B}{3} = kC_A^2 \quad \text{so} \quad r_A = -2kC_A^2 \quad \text{and} \quad r_B = 3kC_A^2$$

  - The numerical value of the rate constant k depends on the how the reaction stoichiometry is written; for example, k is doubled for reaction A → 1.5B compared to 2A → 3B. Obviously r_A etc must be constant regardless.

- For gas-phase reactions over a solid catalyst, we will find the rate law more commonly expressed in terms of the partial pressures of the gases (rather than their concentrations) – and the rate law is often complex (due to adsorption/desorption kinetics –see Section 4).
  - $r = kP_AP_B$ First-order kinetics with respect to both A and B (second order overall)

1.2 Continuous stirred tank reactor (CSTR)

- In an ideal CSTR, the contents are well mixed so that the outlet stream conditions are the same as those <u>everywhere</u> inside the vessel
  - $C_{A,exit} = C_{A,tank}$; $T_{exit} = T_{tank}$



- At steady state, mass flowrate IN = mass flowrate OUT.
- If density is constant, at steady state volumetric flowrate IN = volumetric flowrate OUT (i.e. Q is constant).
  - This is normally a very good approximation for liquid phase reactions (whether for liquid reagents or for reagents dissolved in a solvent).
  - If Q is constant, it is convenient to use concentration as the key variable of interest.
  - The mass balance of species A (IN + MADE = OUT) can be written

    $$QC_{A0} + r_A V = QC_A$$
  - The outlet concentration is

    $$C_A = C_{A0} + r_A \frac{V}{Q} = C_{A0} + r_A \tau \qquad \tau = \text{space time}$$
  - The design volume to achieve a specified concentration is

    $$V = \frac{Q(C_{A0} - C_A)}{-r_A}$$
- If density is changing $Q_{in} \neq Q_{out}$.
  - In that case, it is more convenient to use fractional conversion X_A as the key variable of interest.

    $$X_A = \frac{F_{A0} - F_A}{F_{A0}} = 1 - \frac{F_A}{F_{A0}} = 1 - \frac{Q_{out}C_A}{Q_{in}C_{A0}}$$
    - If Q is constant then $X_A = 1 - C_A/C_{A0}$ and $C_A = C_{A0}(1 - X_A)$, but not otherwise.
  - The mass balance is

    $$F_{A0} + r_A V = F_{A0}(1 - X_A)$$
  - The outlet fractional conversion is

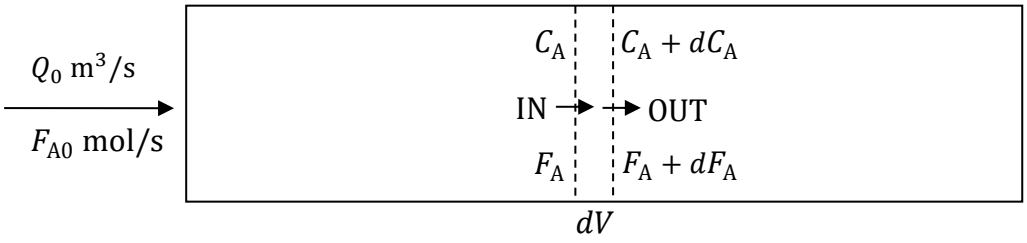
    $$X_A = \frac{-r_A V}{F_{A0}}$$
  - The design volume to achieve a specified conversion is

$$V = \frac{F_{A0} X_A}{-r_A}$$

- If the reactor is not isothermal, an ideal CSTR is still straightforward to analyse because
  - An energy balance can be used to calculate the uniform temperature within the reactor
  - Uniform temperature implies the rate constant is the same everywhere within the reactor.
  - Multiple steady states are possible for non-isothermal CSTRs.

1.3 Plug flow reactor (PFR)

- An ideal PFR has plug flow within it with instantaneous mixing radially, but no mixing axially.
- We perform a mass balance over a volume element



- If density is constant

$$Q C_A + r_A dV = Q(C_A + dC_A)$$

$$dC_A = \frac{r_A}{Q} dV$$

which can be integrated to find concentration along the reactor (including at its outlet).

  - Note that $dC_A = r_A d\tau$ in this case; there is an analogy between reaction in a batch reactor in time t and a PFR with residence time τ if density changes are negligible.

- If density is not constant

$$F_A + r_A dV = F_A + dF_A$$

  - Because $F_A = F_{A0}(1 - X_A)$, we can write $dF_A = -F_{A0} dX_A$, and so

$$r_A dV = -F_{A0} dX_A$$

$$dX_A = \frac{-r_A}{F_{A0}} dV$$

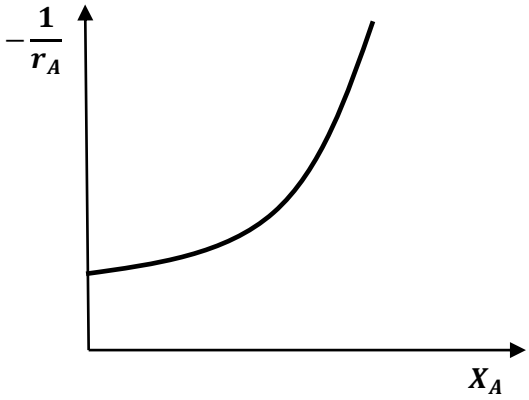
  - The relationship between reactor volume and conversion is therefore

$$V = F_{A0} \int \frac{dX_A}{-r_A}$$

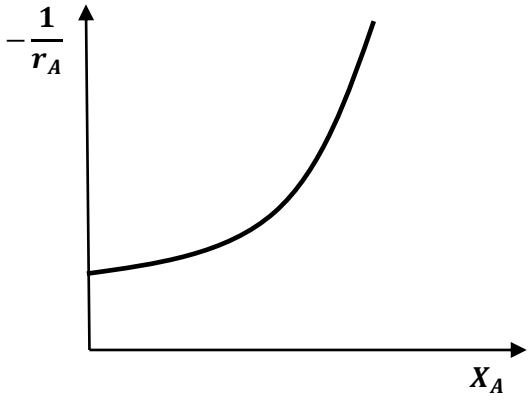
  - To evaluate the integral when density is not constant, the relationship between C_A (in the expression for r_A) and X_A will need to be derived.

PFR vs. CSTR

$$\frac{V}{F_{A0}} = \int \frac{dX_A}{-r_A}$$

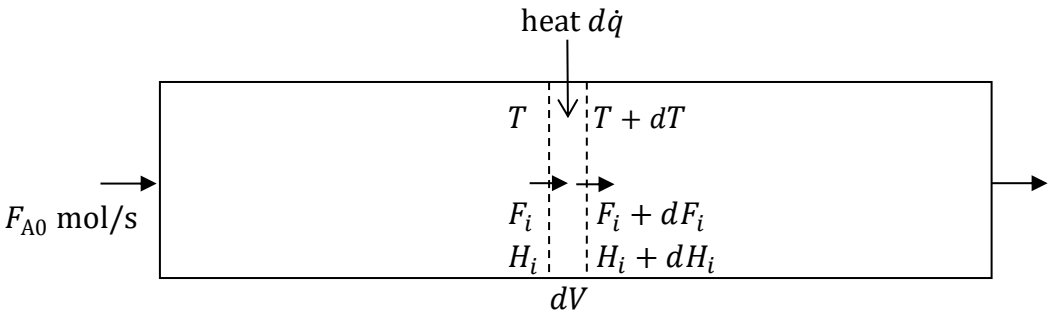


$$\frac{V}{F_{A0}} = \frac{X_A}{-r_A}$$



Non-isothermal PFR

- For a PFR, temperature is likely to vary along the reactor because the reaction will be exothermic or endothermic.
  - We therefore need to perform an energy balance as well as a mass balance.
  - It is likely that temperature will change the density of the fluid; this will certainly be the case for gases.



- Special cases:
  - If adiabatic $d\dot{q}/dV = 0$.
  - An exothermic reaction will have negative $\Delta_r H$ and so increase temperature along the reactor. This is a particularly important case for catalytic reactions.
  - If heat is added (or removed) through the walls of the reactor, then

    $(d\dot{q}/dV) = Ua(T_{hx} - T)$

where a is the surface area per unit volume of the reactor, and T_{hx} is the temperature of the heat exchanging fluid.

- In the general case, the sets of ODEs

$$\frac{dF_i}{dV} = r_i$$

$$\frac{dT}{dV} = \frac{(d\dot{q}/dV) - r\Delta_{\mathrm{r}}H}{\sum F_i C_{\mathrm{P}i}}$$

will need to be solved simultaneously using numerical techniques.

- We also may need to allow for a pressure drop along the reactor

$$\frac{dP}{dV} = \cdots$$

  - Pressure drops can be large for flow of a fluid through a packed bed; this is a common situation in heterogeneous reactors. The pressure drop can be predicted by the Ergun equation (Link: Fluid Mechanics).

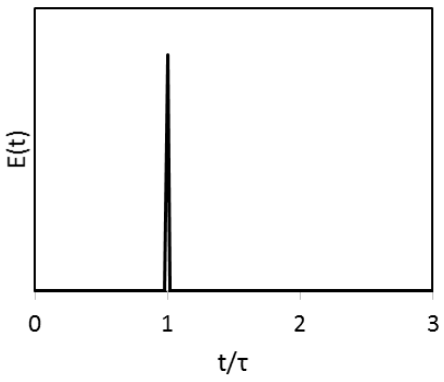
1.4 Non-ideal continuous flow reactors

- Reactors are almost always tubes or tanks, but rarely behave as ideal PFRs or CSTRs.
- We want to be able to size real reactors and be able to predict and control their behaviour.
- Fortunately, we can do this largely on the basis of our knowledge of PFRs/CSTRs.
- The flow pattern is even less likely to be ideal CSTR or ideal PFR for a heterogeneous reactor (the topic of much of this course).

1.4.1 Residence time distribution $E(t)$

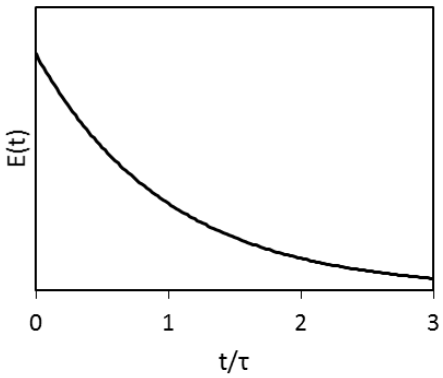
- $E(t)$ is the probability density function for how long molecules spend inside the vessel.
- $E(t)dt$ is the probability that an element of fluid that enters the vessel at time $t = 0$ leaves it between times t and $t + dt$.

- For a PFR, $E(t) = \delta(t - \tau)$

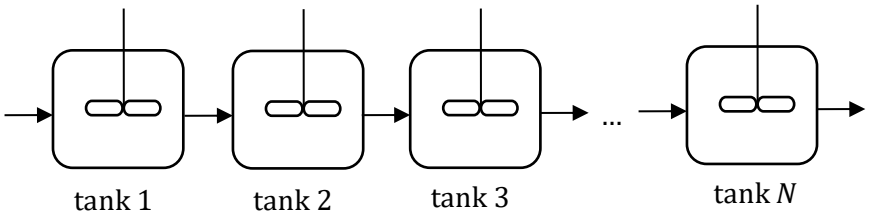


- For a CSTR,

$$E(t) = \frac{1}{\tau}\exp\left(\frac{-t}{\tau}\right)$$



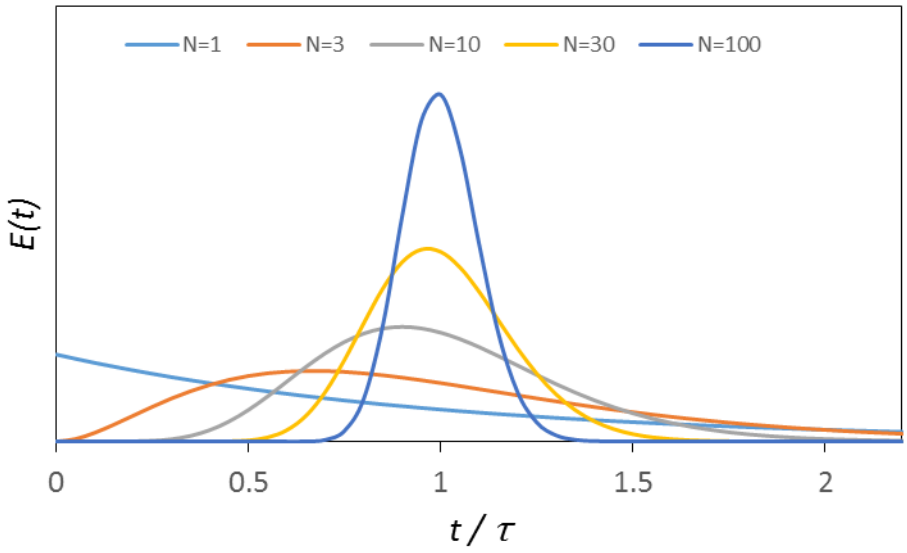
- For N identical CSTRs in series, each with mean residence time τ_i



$$E(t) = \frac{t^{N-1}}{(N-1)!\,\tau_i^N}\exp\left(\frac{-t}{\tau_i}\right)$$

[This result was not given last year, but is useful – no need to remember]

- The overall system will have mean residence time $\tau = N\tau_i$
- The overall variance is $\sigma^2 = \tau^2/N$



- A wide variety of reactors can be modelled as a chain of N CSTRs

1.4.2 Micromixing and Macromixing

- We want to predict conversion in reactors with different flow patterns.
- We can measure experimentally the RTD for the reactor.
- Note that the flow pattern specifies the RTD…. but the RTD does not specify the flow pattern.
  - For instance, the reactor sequence PFR + CSTR has the same RTD as CSTR + PFR, but the flow pattern is different.
  - The PFR + CSTR flow pattern does not necessarily give the same conversion as the CSTR + PFR flow pattern.
- We need to consider the state of mixing in the vessels as well as RTD.

Micromixing

- In the limit of perfect micromixing, we can apply a model of "maximum mixedness" in which there is good mixing at the molecular level.
- Design equations for CSTR and PFR (Sections 1.2 and 1.3) can be used.

Macromixing

- In the limit of perfect macromixing, we can apply a model of "complete segregation".
  - Once inside the vessel, the complete segregation model assumes perfect mixing between molecules within the globule, but no mixing between the globule and fluid outside the globule.
  - We can therefore treat each globule as a mini-batch reactor. The conversion in each globule will be $X_{\text{batch}}(t)$.
- The average conversion in the reactor with macromixing will therefore be

$$\overline{X} = \int_0^\infty X_{\text{batch}}(t)\, E(t)\, dt$$